ABSTRACT

Blind spots surrounding vehicles present a safety concern because nearby objects may be difficult to observe directly. This project develops an ESP32-based tabletop prototype that demonstrates three-zone proximity monitoring in the left, right and rear directions. Three HC-SR04 ultrasonic sensors measure object distances, while two HLK-LD2410C mmWave modules are connected to independent hardware UART interfaces for communication-status monitoring. A 10 kΩ potentiometer produces a simulated speed between 0 and 100 km/h. The programmed decision logic applies one of three short-range warning bands based on that speed: 5/10 cm (danger/warning) at low speed, 6/12 cm at medium speed and 8/15 cm at high speed. These values are intentionally chosen for a small stationary model rather than real-road operation.

Measurements are acquired sequentially to reduce interference among the ultrasonic modules and stabilized using a three-sample median filter. The controller assigns each zone a SAFE, WARNING, DANGER or NO ECHO condition, prioritizes the highest observed risk, drives red/yellow/green LED indicators and a buzzer with distinct alert patterns, and updates a 128×64 SSD1306 OLED display. In the final supplied firmware, an onboard Wi-Fi access point hosts a local monitoring page and JSON API. The OLED initially presents the network name, password and IP address; once the browser begins requesting dashboard data, the display returns to live monitoring.

The working prototype demonstrates distance-based warnings, speed-adaptive thresholds and local visualization. The current firmware monitors radar UART availability but does not use radar target distance/presence values in its hazard computation; accordingly, radar–ultrasonic measurement fusion is identified as future work rather than an implemented result. The prototype is intended for laboratory education and demonstration, not certified automotive collision avoidance.

Keywords: ESP32, blind-spot monitoring, HC-SR04, LD2410C, ultrasonic ranging, speed-adaptive alert, SSD1306 OLED, Wi-Fi dashboard.

CONTENTS

CHAPTER 1 INTRODUCTION

CHAPTER 2 SYSTEM DESIGN AND OPERATION

CHAPTER 3 HARDWARE DESIGN AND DESCRIPTION

CHAPTER 4 HARDWARE IMPLEMENTATION

CHAPTER 5 SOFTWARE DESCRIPTION

CHAPTER 6 SOFTWARE IMPLEMENTATION

CHAPTER 7 EXPERIMENTAL RESULTS

CHAPTER 8 APPLICATIONS, ADVANTAGES AND DISADVANTAGES

CHAPTER 9 CONCLUSION AND FUTURE SCOPE

REFERENCES

APPENDIX A - GPIO CONNECTION SUMMARY

APPENDIX B - DEMONSTRATION AND VALIDATION CHECKLIST

LIST OF FIGURES

Figure 2.1 Functional system block diagram

Figure 2.2 Conceptual plywood mounting arrangement

Figure 4.1 Detailed ESP32 GPIO connection diagram

Figure 6.1 Software implementation flowchart

Figure 7.1 Illustrative local dashboard preview

CHAPTER 1

INTRODUCTION

1.1 Overview

A vehicle blind spot is a region around the vehicle that may not be readily visible to the driver. Electronic proximity-warning systems can supplement visual awareness by monitoring surrounding regions and warning when objects approach a defined distance. At prototype scale, ultrasonic sensors provide a simple method for evaluating range-sensitive alerts without requiring a moving vehicle. This project implements a left, right and rear monitoring arrangement on a stationary plywood board with a toy car representing the host vehicle.

The central ESP32 microcontroller acquires three ultrasonic distances, monitors two connected 24 GHz mmWave modules over separate UARTs, samples a vehicle-speed potentiometer and calculates the appropriate warning state. Feedback is provided through an SSD1306 OLED, three LEDs, an active buzzer and a browser dashboard that the ESP32 serves over its own Wi-Fi access point. The design emphasizes visible operation, repeatable short-distance demonstrations and low-cost readily available components.

1.2 Problem statement

Small educational vehicle models generally lack a straightforward method to indicate whether a toy obstacle is present to the left, right or rear. A single fixed proximity threshold also does not demonstrate how a warning policy might change with vehicle speed. The problem addressed is therefore the design of an affordable, modular embedded demonstrator that measures obstacle distances in three zones, indicates a clear priority-based warning, allows adjustable simulated speed, and presents live feedback locally without cloud services.

1.3 Objectives

- Develop a three-zone ultrasonic proximity monitoring arrangement for left, right and rear regions.
- Read and maintain serial communication with two HLK-LD2410C modules on independent ESP32 UART ports.
- Reduce short-term distance fluctuations using a three-sample median filter and sensor timeouts.
- Apply three user-defined short-range distance threshold bands selected by a 10 kΩ speed potentiometer.
- Provide status feedback through LEDs, an audible buzzer, the OLED and the Serial Monitor.
- Host a locally accessible read-only Wi-Fi dashboard and present connection instructions on the OLED.
- Demonstrate normal, warning, danger and unavailable-sensor cases without claiming road-use suitability.

1.4 Literature survey and technical background

At the component and system level, the design draws on four established technology areas rather than reproducing any one prior research model. The ESP32 platform provides digital GPIO, hardware UART, ADC, I²C and Wi-Fi capabilities [1]. Ultrasonic ranging estimates distance from echo propagation time [2]. The HLK-LD2410 series provides 24 GHz presence-sensing information through a serial interface and is commonly used for human-presence applications [3]. OLED interface libraries and local browser interfaces offer compact human-machine monitoring [4, 5].

Technology / approach	Contribution to this prototype	Current constraint
Ultrasonic time-of-flight sensing	Direct short-range measurements in three physical directions	Reflections, close-range blind region and transducer geometry affect measurements
24 GHz LD2410C presence radar	Two modules connected with UART link-state reporting	Not validated for small toy-car classification; target values are not fused
ESP32 embedded control	Sensor acquisition, threshold logic and multiple outputs	Demo firmware is not safety certified or fault tolerant
Local Wi-Fi monitoring	Live browser access to range, speed, status and historical graph	Only accessible to devices on the ESP32 access-point network

1.5 Existing and proposed system summary

A simple existing educational demonstration often uses a single ultrasonic sensor and an LED that switches at a fixed distance. Such a setup offers limited spatial information and does not illustrate changing warning limits. The proposed system extends the basic arrangement to three independently measured directions, derives a single priority-based overall risk status, and varies programmed danger and warning limits according to a simulated speed setting. It also presents the same conditions on multiple output channels so that physical and digital responses can be observed together.

1.6 Hardware and software components used

Hardware component	Quantity	Purpose
ESP32 DevKit V1	1	Main controller and Wi-Fi access point
HC-SR04 ultrasonic sensor	3	Left, right, rear distance measurements
HLK-LD2410C mmWave module	2	Radar UART communications
SSD1306 I²C OLED, 128×64	1	Local visual feedback
Green, yellow and red LEDs	3	Overall status indicators
Buzzer module	1	Audible warnings
10 kΩ linear potentiometer	1	Simulated vehicle speed input
220 Ω LED resistors; echo dividers	As required	Current limiting and voltage-level protection
Plywood, holders, jumper wires, USB supply	As required	Stationary model and power distribution

Software: Arduino IDE with ESP32 board support; Arduino C/C++; Wire, Adafruit_GFX and Adafruit_SSD1306 for the OLED; ld2410 library for radar UART; WiFi and WebServer for the locally hosted dashboard; and a web interface implemented using HTML, CSS and JavaScript. The final code is a single sketch with embedded dashboard resources.

1.7 Organization of the report

Chapter 2 explains the functional architecture and warning policy. Chapter 3 describes the hardware. Chapter 4 presents connections and assembly. Chapter 5 introduces development tools and interfaces. Chapter 6 describes the firmware algorithm. Chapter 7 documents qualitative operation and validation scenarios. Chapter 8 summarizes applications and limitations. Chapter 9 provides conclusions and future scope.

1.8 Summary

This chapter defined the educational need, system scope, component set and main objectives. The following chapters describe the implemented logic and circuitry in terms consistent with the supplied final program.

CHAPTER 2

SYSTEM DESIGN AND OPERATION

2.1 Block diagram

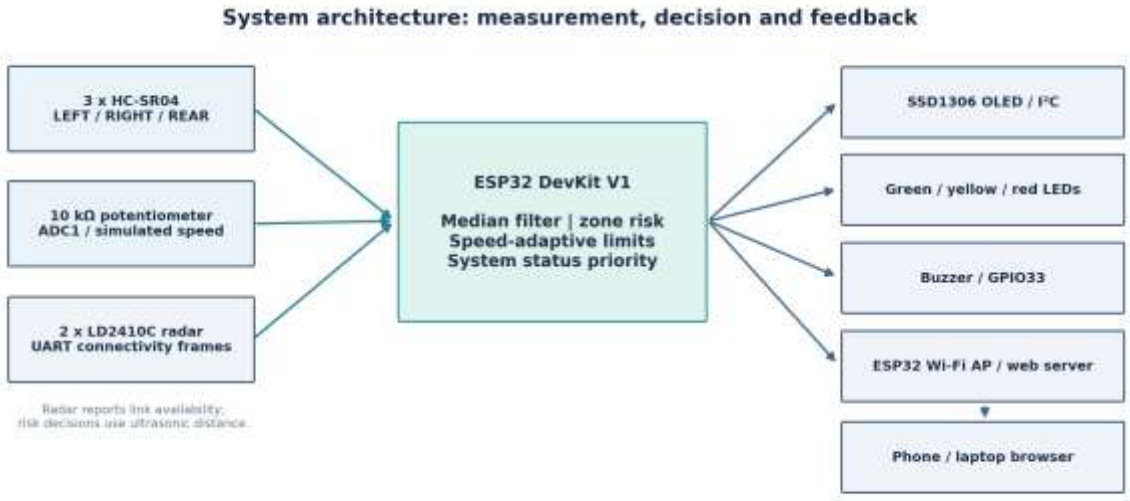


Figure 2.1 Functional block diagram of the ESP32-based blind-spot prototype

2.2 Working mechanism

The three HC-SR04 sensors are triggered one at a time. For each sensor, the ESP32 generates a short TRIG pulse and measures the HIGH time of the corresponding ECHO signal with a 25,000 µs timeout. The estimated distance is calculated using the approximate speed of sound, d = (t × 0.0343 cm/µs) / 2. Valid measurements from 2 cm to 400 cm enter a circular buffer of up to three samples. The median of the available readings becomes the reported distance; after three consecutive invalid echoes, the corresponding zone is marked NO ECHO.

The two LD2410C modules stream radar communication frames at 256,000 baud to separate ESP32 hardware UART channels. The program repeatedly calls each module's read() method, then displays the connectivity state from isConnected(). These link checks are independent of the ultrasonic warning algorithm. They do not establish that toy-car obstacles have been reliably recognized by radar.

The 10 kΩ potentiometer is connected to ESP32 ADC1 GPIO36 and produces a simulated speed between 0 and 100 km/h. The ADC value is averaged over eight reads and a lightweight smoothing formula is applied. One of three threshold bands is selected. For each zone, the code first checks DANGER, then WARNING, then SAFE; a missing ultrasonic result becomes NO ECHO. Across all zones, DANGER has highest priority, followed by WARNING, then CHECK SENSOR, then SAFE.

2.3 Speed-adaptive prototype thresholds

Speed range from potentiometer	DANGER when distance is	WARNING when distance is	SAFE when distance is

0–30 km/h	≤ 5 cm	> 5 cm and ≤ 10 cm	> 10 cm
31–60 km/h	≤ 6 cm	> 6 cm and ≤ 12 cm	> 12 cm
61–100 km/h	≤ 8 cm	> 8 cm and ≤ 15 cm	> 15 cm

These are deliberately short-range tabletop demonstration values. All three directions apply the same thresholds at a given simulated speed. The simulated numerical speed is not measured from a moving vehicle and must not be presented as real speed estimation.

2.4 Overall warning indicators

Overall state	LED output	Buzzer pattern	OLED / browser indication
SAFE	Green steady	OFF	SAFE
WARNING	Yellow steady	150 ms ON / 850 ms OFF	WARNING
DANGER	Red steady	150 ms ON / 150 ms OFF	DANGER
CHECK SENSOR	Yellow blinks every 500 ms	80 ms ON every 3 s	CHECK or CHECK SENSOR

2.5 Wi-Fi and display operation

The ESP32 starts a local access point identified in the source as BlindSpot-Pro, with the access-point IP normally 192.168.4.1. The main OLED screen initially shows the SSID, password and IP address, instructing the user to open the address in a browser. The HTTP server responds to "/" with the dashboard page and "/api/status" with live JSON sensor and system information. Receiving a dashboard page or API request marks the browser as active. While requests continue, the OLED shows the original live sensor screen; after approximately 12 seconds without browser requests, it returns to connection instructions.

The web dashboard is read-only and does not alter warning distances or output states. It is a local network interface, not a cloud service. Phone or laptop access requires joining the ESP32's Wi-Fi access point.

2.6 Physical arrangement

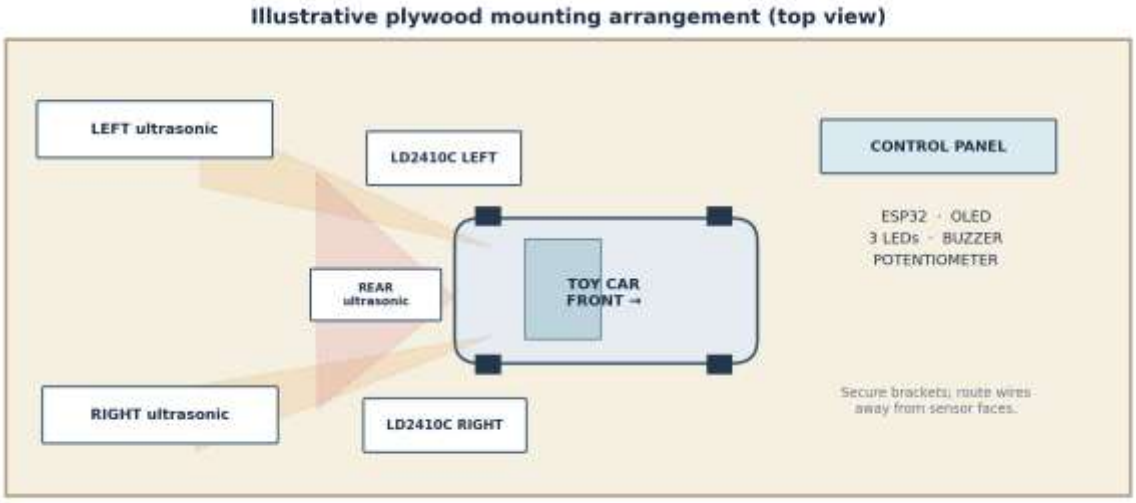


Figure 2.2 Conceptual top view of the stationary plywood demonstrator

The toy vehicle provides a visual reference for three detection zones. Each ultrasonic sensor is mounted on a short stand with the two transducers facing the region being tested. The electronics panel is separated from the active detection area to avoid accidental sensor obstruction. Modules are secured mechanically and their cable paths are routed along the board edges.

CHAPTER 3

HARDWARE DESIGN AND DESCRIPTION

3.1 ESP32 DevKit V1 microcontroller

The ESP32 DevKit V1 provides the application's processing and communication resources. GPIO pins operate at 3.3 V logic levels; they must not be connected directly to 5 V ECHO outputs. Hardware Serial2 is mapped to GPIO16/17 and Hardware Serial1 to GPIO26/27. The I²C pins GPIO21 and GPIO22 communicate with the SSD1306. ADC1 GPIO36 receives the speed potentiometer, while remaining GPIO pins operate the ultrasound TRIG channels, indicator LEDs and buzzer. The built-in Wi-Fi capability hosts the monitoring interface. The USB interface is used to program the board and can supply power during a controlled tabletop demonstration.

3.2 HLK-LD2410C mmWave radar modules

Two HLK-LD2410C 24 GHz FMCW radar presence modules are installed at left and right positions. The final firmware powers them separately from the logic signal path, initializes them at 256,000 baud and maintains a communications check through the ld2410 library. UART TX of each module goes to the matching ESP32 RX pin, and module RX goes to ESP32 TX. A shared ground is required. These modules are primarily intended for presence sensing, including human movement; their target measurements should not be assumed to provide reliable centimetre-level ranging of small toy cars without independent validation.

3.3 HC-SR04 ultrasonic distance sensors

Each HC-SR04 sends an ultrasonic burst when its TRIG input receives a pulse. The duration of the ECHO signal is proportional to the return travel time. Three modules provide left, right and rear ranges. The firmware uses a 65 ms interval between consecutive sensor triggers and cycles through the three zones, reducing simultaneous acoustic interference. Each ECHO pin is connected through a resistor divider so that a 5 V-level pulse does not exceed the ESP32 GPIO input limits. Very close objects, angled targets and acoustically absorbent materials can reduce measurement quality.

3.4 SSD1306 OLED display

The 128×64 monochrome OLED operates on the I²C bus at address 0x3C using SDA GPIO21 and SCL GPIO22. It has two display states: a network-connection instruction screen and a live monitoring screen. The latter shows three distances and their classifications, a compact radar link line, the simulated speed and the overall status. Its 250 ms refresh interval is independent of the Serial Monitor interval.

3.5 Three indicator LEDs

The green LED on GPIO23 represents SAFE, the yellow LED on GPIO19 represents WARNING and the red LED on GPIO25 represents DANGER. The LEDs are connected through current-limiting resistors. When no valid zone measurement is available and no higher-priority warning

exists, the yellow LED blinks as a diagnostic indication. Only one steady risk LED should be active at a time.

3.6 Buzzer module

A buzzer connected to GPIO33 provides audible differentiation between warning levels. The firmware uses millis()-based periodic patterns, avoiding long deliberate delay calls for alert timing. The installed code assumes an active-high control signal, which must match the actual buzzer module. For a bare/high-current buzzer, an appropriate transistor driver circuit is needed rather than connecting the load directly to an ESP32 GPIO.

3.7 Vehicle-speed potentiometer

A 10 kΩ linear potentiometer has its two outer terminals connected to 3.3 V and GND; the center wiper connects to ADC1 GPIO36. The analog input is converted to a nominal 0–100 km/h range. The selected threshold band changes only when the smoothed reading crosses 30 km/h or 60 km/h. This potentiometer emulates the speed input that a future real system might obtain from a validated wheel or vehicle-speed data source.

3.8 Power and protection

The prototype needs stable logic and sensor power with a common ground reference. The HC-SR04 modules and LD2410C modules normally require 5 V supply connections; the OLED and ESP32 GPIO signals use 3.3 V logic as appropriate for their actual breakout boards. A regulated, adequately rated supply or suitable USB power arrangement should be used. Three ECHO resistor-divider networks protect GPIO34, GPIO35 and GPIO32. These components should be tested for voltage and ground continuity before connecting to the ESP32.

3.9 Mechanical mounting materials

The stationary assembly uses plywood, a centrally positioned toy car, upright brackets or acrylic/printed mounts for the range sensors, a rear or side electronics panel, mounting screws or standoffs, insulated hookup wire and cable ties. Sensor faces must remain unobstructed. The potentiometer should be accessible for a demonstration while cables and connectors remain firmly secured.

CHAPTER 4

HARDWARE IMPLEMENTATION

4.1 Circuit connections and wiring diagram

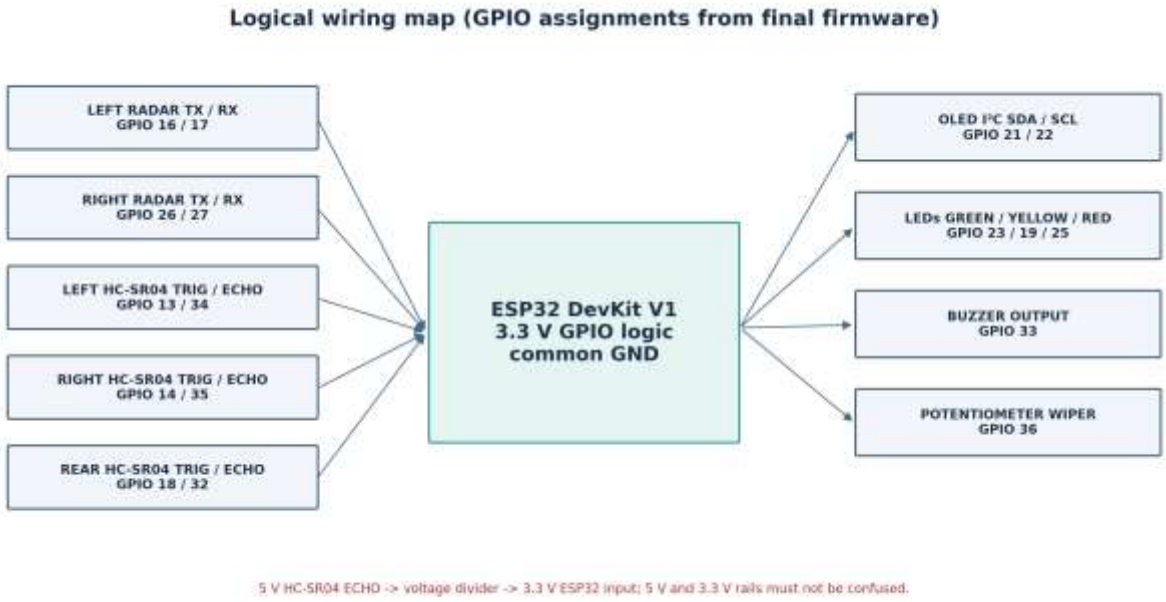


Figure 4.1 Detailed GPIO assignment diagram derived from the final sketch

4.2 Pin description

Module / component	ESP32 pin(s)	Interface / purpose
Left LD2410C	16 (RX), 17 (TX)	UART2 at 256000 baud
Right LD2410C	26 (RX), 27 (TX)	UART1 at 256000 baud
Left HC-SR04	13 TRIG, 34 ECHO	Digital trigger + level-shifted input
Right HC-SR04	14 TRIG, 35 ECHO	Digital trigger + level-shifted input
Rear HC-SR04	18 TRIG, 32 ECHO	Digital trigger + level-shifted input
SSD1306 OLED	21 SDA, 22 SCL	I²C, address 0x3C
Green LED	23	GPIO output through resistor
Yellow LED	19	GPIO output through resistor
Red LED	25	GPIO output through resistor
Buzzer	33	Digital output
Speed potentiometer	36	ADC1 input, 0–3.3 V

4.3 Echo-level protection

The HC-SR04 ECHO lines must not be connected directly to the ESP32 because a typical ECHO output can reach 5 V. A two-resistor divider may be wired with an upper resistor of approximately 1 kΩ from ECHO to the ESP32 input and a lower resistor of approximately 2 kΩ from input to ground, yielding about 3.3 V from a 5 V pulse. Use the same divider scheme on left, right and rear

inputs. Confirm the actual resistor placement and measure the output with a meter before connecting it to the microcontroller.

4.4 Assembly method

1. Mark the intended left, right and rear positions on the plywood and fix the toy car in the center.
2. Mount three ultrasonic modules on stands, orienting each transducer pair toward its demonstration zone.
3. Mount the two LD2410C modules and the controller panel, ensuring nearby wiring does not obscure the antenna faces.
4. Secure the OLED, LED panel, buzzer and speed potentiometer where they are visible and accessible.
5. Wire regulated power and common grounds; verify resistor dividers on all three ultrasound ECHO inputs.
6. Connect the GPIO signal wires according to Table 4.1 and check for shorts and loose jumpers.
7. Upload firmware, confirm boot messages and verify each sensor zone one at a time before running the full demonstration.

4.5 Implementation precautions

The detection ranges in this project are intentionally set to very small values. Allow sufficient physical clearance for objects to enter the 4–15 cm demonstration region without striking the sensor. In a tabletop environment, the board edges, mounting brackets or the toy car itself may be reflected by ultrasonic sensors and should be repositioned if they cause persistent unintended warnings. The system is stationary and should not be installed as an operational vehicle blind-spot safety system.

CHAPTER 5

SOFTWARE DESCRIPTION

5.1 Arduino IDE

Arduino IDE is used to edit, verify and upload the C/C++ sketch to the ESP32 DevKit. The ESP32 board package supplies GPIO, timing, UART, ADC and Wi-Fi APIs. A serial terminal at 115200 baud displays initialization messages, individual zone measurements and the overall status. The final sketch contains one setup() function for initialization and a continuous loop() for polling, timing and output updates.

5.2 Required libraries

Library	Role in final firmware
Arduino.h	Arduino/ESP32 core, timing and GPIO
ld2410.h	Two radar UART module objects and isConnected() reporting
Wire.h	I²C bus to SSD1306
Adafruit_GFX.h	Text and graphic drawing primitives
Adafruit_SSD1306.h	OLED buffer and screen update
WiFi.h	Local Wi-Fi access-point mode
WebServer.h	HTTP page and JSON status endpoints
math.h / string.h	Invalid-distance handling and status comparison

5.3 Communication interfaces

UART2 and UART1 communicate with the two radar modules at 256000 baud, while the USB Serial Monitor prints at 115200 baud. The OLED uses I²C on GPIO21/22. Ultrasonic sensors use discrete trigger/echo GPIO and the potentiometer uses ADC1. The web server listens on HTTP port 80. Because the ESP32 generates its own access point, a cloud account or external router is not necessary.

5.4 Firmware scheduling

Operation	Programmed timing
Ultrasonic ping	65 ms between sequential sensor triggers
Ultrasonic ECHO timeout	25000 µs per measurement
Potentiometer update	100 ms
OLED update	250 ms
Serial Monitor report	750 ms
Dashboard JSON polling	Approximately 700 ms in the browser
OLED browser inactivity timeout	12000 ms

The program uses elapsed-time checks rather than pausing the entire loop for each display, LED or buzzer event. A pulseIn() operation can still block for up to its specified timeout if no ultrasonic echo arrives; the selected timeout bounds that delay.

5.5 Web dashboard description

The dashboard is embedded as a PROGMEM HTML string, allowing the ESP32 to serve a full single-page interface from its own flash. The browser requests "/api/status" periodically, and the response contains simulated speed, active warning thresholds, overall status, two radar connection booleans, three zone distances and statuses, uptime and access-point address. JavaScript plots recent range data and displays risk transitions. The user cannot modify detection parameters from the browser in this version.

CHAPTER 6

SOFTWARE IMPLEMENTATION

6.1 Software flowchart

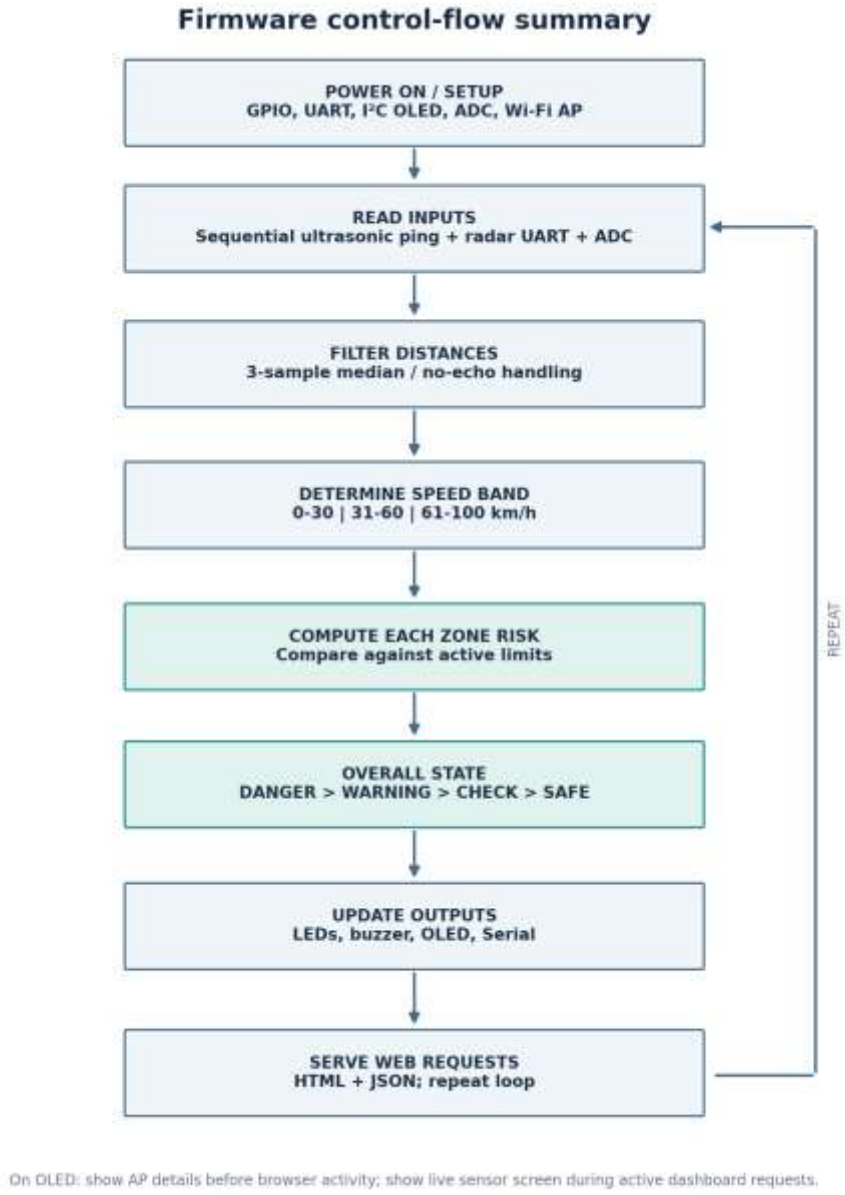


Figure 6.1 Overall software algorithm and output-update sequence

6.2 Algorithm

1. Start the ESP32 and configure USB Serial output at 115200 baud.
2. Initialize ADC, LED outputs, buzzer GPIO and three ultrasonic TRIG/ECHO pairs.
3. Start the OLED over I²C and initialize two radar hardware UART channels at 256000 baud.
4. Create the local Wi-Fi access point and HTTP routes; show Wi-Fi details on the OLED.
5. Read the two radar communication streams frequently to update link-state information.

6. At each scheduled ping event, trigger only the active ultrasonic sensor; measure its ECHO pulse with timeout.
7. Convert a valid pulse to centimetres and update the three-sample median for that zone; classify missing echoes as unavailable after repeated failures.
8. Sample the speed potentiometer, smooth the reading and select low, medium or high simulated-speed warning limits.
9. For each zone, classify distance as DANGER, WARNING, SAFE or NO ECHO.
10. Determine the overall state using priority DANGER > WARNING > CHECK SENSOR > SAFE.
11. Update three LED indicators and non-blocking buzzer patterns based on the overall state.
12. Update OLED network instructions or live monitoring depending on recent browser requests.
13. Print UART connectivity, simulated speed, distance limits and zone states to the Serial Monitor.
14. Serve web-page requests and return current data through JSON; repeat steps 5–14 continuously.

6.3 Distance filtering and validity

For each sensor zone, the code stores up to three valid distance samples in a circular array. With one valid sample the filtered output is that value; with two samples it is their average; with three samples it is the median after sorting. Filtering reduces the effect of one isolated outlier but also introduces a short response delay while the buffer fills. The code resets the sample buffer if three consecutive reads are invalid, resulting in NO ECHO rather than a falsely definite safe distance.

6.4 Risk classification logic

The active thresholds are returned by dangerLimit() and warningLimit(). For any valid distance d, the classification is DANGER if d <= dangerLimit(), WARNING if d <= warningLimit(), otherwise SAFE. If distance is not a valid number, the class is NO ECHO. The overall status checks all three zones and retains the most severe applicable condition. Consequently, a danger alert in any single direction overrides a safe result in either of the remaining zones.

6.5 Example threshold source extract

```
const float DANGER_DISTANCE = 5.0f;
const float WARNING_DISTANCE = 10.0f;

float dangerLimit() {
  if (speedKmh > 60) return 8.0f;
  if (speedKmh > 30) return 6.0f;
  return DANGER_DISTANCE;
}

float warningLimit() {
  if (speedKmh > 60) return 15.0f;
  if (speedKmh > 30) return 12.0f;
  return WARNING_DISTANCE;
}
```

6.6 Live data and OLED selection

Two HTTP handlers supply dashboard content: the root route returns HTML, CSS and JavaScript, and /api/status returns JSON. Each successful page/status request updates a timestamp. Until the first dashboard request, or after a 12-second inactivity period, the OLED displays the Wi-Fi name and current access-point IP. During an active browser session, it displays live monitoring. Both OLED screens are informational; neither changes the underlying sensor sampling or warning policy.

6.7 Error handling

If one or more ultrasonic zones have no valid measurements, the program reports NO ECHO for those zones. If no valid zone is in a warning/danger range, the overall result is CHECK SENSOR and the yellow LED and buzzer use diagnostic patterns. Failure to initialize the OLED is reported over Serial without preventing the remainder of the system from operating. The firmware also shows radar OFFLINE when communication frames are not received. Further fault tolerance, such as sensor redundancy and independent watchdog behavior, would be needed for any safety-critical application.

CHAPTER 7

EXPERIMENTAL RESULTS

7.1 Prototype verification approach

The supplied project history describes the assembled prototype as working, including obstacle sensing, LED and buzzer alerts, OLED monitoring, simulated speed control and the access-point dashboard. The following tables summarize the implemented test scenarios and the expected states according to the final firmware. They are representative demonstration cases, not a claim of laboratory calibration, certified accuracy or time-series measurements. No complete raw experimental log or metrology dataset was supplied for quantitative error analysis.

7.2 Low-speed obstacle-response scenarios

Zone under test	Example observed/placed distance	Expected zone state	Overall output if other zones safe
LEFT	15 cm	SAFE	Green LED; silent
LEFT	8 cm	WARNING	Yellow LED; slow beep
LEFT	4 cm	DANGER	Red LED; fast beep
RIGHT	9 cm	WARNING	Yellow LED; slow beep
REAR	4 cm	DANGER	Red LED; fast beep

All cases in Table 7.1 assume the potentiometer indicates 0–30 km/h and the other directions read SAFE. For real demonstrations, the displayed distance may differ slightly from a ruler reading because sensor alignment, reflected sound and median filtering affect the result.

7.3 Speed adjustment at fixed obstacle distance

Simulated speed band	Fixed sample distance	Active danger limit	Expected classification
0–30 km/h	7 cm	5 cm	WARNING
31–60 km/h	7 cm	6 cm	WARNING
61–100 km/h	7 cm	8 cm	DANGER

The comparison demonstrates that the risk classification can change without moving the obstacle. It is the programmed speed-dependent threshold, rather than any new distance measurement, that changes the classification in this scenario.

7.4 OLED and local dashboard behavior

Before a browser loads the dashboard, the OLED presents access-point instructions. A user connects to BlindSpot-Pro and opens the printed IP address in a web browser. When the dashboard requests live data, the display switches to the original sensor view. The dashboard shows zone distance cards, speed gauge, thresholds, overall state, two radar communication indicators, rolling distance graphs and recent risk transitions. If the browser stops requesting data for approximately 12 seconds, the OLED returns to network instructions.



Figure 7.1 Illustrative dashboard UI preview; values shown are sample data, not measured experimental results

7.5 Functional test matrix

Function	Method of checking	Expected result
Left/right/rear sensing	Move toy target separately toward each sensor	Only the relevant zone distance changes materially
Danger priority	Place object within current danger limit	Overall state DANGER and red LED/fast beep
Warning indication	Place object in warning band	Yellow LED and slower beep
Speed control	Turn GPIO36 potentiometer	Displayed simulated speed and active thresholds update
Missing echo	Block/remove echo or disconnect a sensor safely	NO ECHO / CHECK SENSOR according to overall priority
Radar UART	Keep/remove UART communication under controlled test	ONLINE/OFFLINE communication status changes
Wi-Fi dashboard	Join AP and browse IP	Web page opens and displays refreshed data

OLED mode switch	Open then close/idle browser	IP display changes to live view and returns after idle

7.6 Discussion and limitations of evaluation

The programmed warning values of 5 cm and 10 cm are near the lower end of the useful range of a commodity HC-SR04. Performance depends on the sensor, surface and environment; therefore, physical testing at each threshold is essential. The program uses 65 ms between ultrasonic pings and three-sample filtering rather than simultaneous measurements. These are sound design choices for a tabletop model, but they do not establish automotive reaction-time performance.

It is important to separate the measured and monitored functions: ultrasonic sensors deliver the distances used to compute risk; the radar modules deliver UART connectivity state. Although both sensor technologies are physically integrated, the present firmware does not mathematically fuse their target measurements. Accordingly, a measured improvement in detection accuracy due to multi-sensor data fusion cannot be claimed from this implementation. This boundary is explicitly retained in the conclusions and future work.

CHAPTER 8

APPLICATIONS, ADVANTAGES AND DISADVANTAGES

8.1 Applications

- Automotive electronics laboratory demonstrations of left/right/rear proximity zones.
- Embedded-systems coursework involving GPIO, ADC, UART, I²C and local HTTP services.
- Training on multi-level hazard classification and speed-sensitive threshold policies.
- Science exhibitions and prototype showcases using toy vehicles and plywood mounting.
- Educational evaluation of ultrasonic measurements, noise filtering and missing-sensor behavior.

8.2 Advantages

- Three separately monitored ultrasonic directions provide greater spatial coverage than a one-sensor model.
- Clear multi-channel feedback through OLED, LEDs, buzzer, Serial Monitor and browser interface.
- A potentiometer makes the speed-policy effect visible without a moving vehicle.
- A local ESP32 access point allows browser monitoring without external internet connectivity.
- Modular wiring and low-cost components support classroom assembly and repair.
- Sequential triggering and median filtering improve interpretability of noisy echo measurements.

8.3 Disadvantages and limitations

- The system is a stationary short-range demonstration, not a validated road-use blind-spot system.
- Ultrasonic returns depend on target angle, surface characteristics, beam width and interference.
- Very small distance thresholds of 5–15 cm require careful sensor positioning and close-range validation.
- Radar target measurements are not currently fused into the warning algorithm; UART link status alone is not object detection.
- The simulated speed is manually set; no wheel, GPS or CAN-bus vehicle-speed input is implemented.
- The Wi-Fi dashboard depends on access-point coverage and does not provide security suitable for public deployment.
- No calibrated accuracy, false-alarm rate, environmental robustness or end-to-end latency measurements were supplied.

CHAPTER 9

CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

An ESP32-based Intelligent Blind-Spot Detection System was implemented as a stationary three-zone tabletop prototype. The completed firmware reads left, right and rear ultrasonic distances, applies three-sample median filtering, calculates a speed-dependent warning state and provides immediate visual and audible indications through a 128×64 OLED, three LEDs and a buzzer. A 10 kΩ potentiometer selects among low-, medium- and high-speed demonstration policies. The final version also hosts a local Wi-Fi dashboard, with the OLED automatically switching between connection instructions and live measurements according to browser activity.

The project demonstrates the interaction of several embedded-system interfaces and a practical feedback workflow at low cost. Two LD2410C mmWave modules are connected and their UART connectivity is monitored; however, radar target values are not currently part of risk classification. The central verified functionality is therefore speed-adaptive ultrasonic proximity alerting with supplementary radar communication monitoring. The model is an educational demonstration and is not a vehicle-ready collision-avoidance product.

9.2 Future scope

1. Validate radar target readings with controlled object sizes and distances; only then develop and quantify true radar–ultrasonic fusion.
2. Replace the potentiometer with a validated speed source such as wheel sensing or vehicle CAN data.
3. Perform repeated distance-reference tests and quantify range error, delay and false alarms.
4. Extend to longer automotive-relevant distances and compensate for changing environmental conditions.
5. Improve enclosure design, sensor field-of-view placement, electrical protection and vibration resistance.
6. Add nonvolatile logging or time-stamped data export for systematic test analysis.
7. Use authentication and other security measures if extending browser monitoring beyond a closed classroom access point.
8. Implement diagnostic watchdogs and fault-state handling for sensor/power communication failures.

REFERENCES

[1] Espressif Systems, ESP32 Series Datasheet (official technical documentation). https://documentation.espressif.com/esp32_datasheet_en.html

[2] HC-SR04 Ultrasonic Ranging Module, device operating principle and interface documentation supplied with the module (TRIG/ECHO timing and supply requirements).

[3] N. Reynolds, ld2410: Arduino library for Hi-Link 24 GHz FMCW radar sensors. https://github.com/ncmreynolds/ld2410

[4] Adafruit Learning System, Monochrome OLED Breakouts - Arduino Library and Examples. https://learn.adafruit.com/monochrome-oled-breakouts/arduino-library-and-examples

[5] Arduino, Arduino IDE and ESP32-compatible sketch development environment. https://www.arduino.cc/en/software

[6] Project source: final supplied ESP32 firmware, dated 31 October 2026; includes GPIO definitions, short-range threshold policy, OLED mode selection and embedded Wi-Fi dashboard HTML.

APPENDIX A - GPIO CONNECTION SUMMARY

Subsystem	Pins / electrical note
Left LD2410C	TX -> ESP32 RX16; RX <- TX17; 5 V supply and common GND
Right LD2410C	TX -> ESP32 RX26; RX <- TX27; 5 V supply and common GND
Left HC-SR04	TRIG13, ECHO34 through divider
Right HC-SR04	TRIG14, ECHO35 through divider
Rear HC-SR04	TRIG18, ECHO32 through divider
OLED SSD1306	SDA21, SCL22, 0x3C
LEDs	Green23, yellow19, red25 through current-limiting resistors
Buzzer	GPIO33, polarity/module rating as required
Potentiometer	Wiper GPIO36; outer pins to 3.3 V and GND

Power and wiring note: The wiring table reports logical connections described in the source and assembly discussion. Check the pin labels of the exact module/breakout boards before soldering. Any external power rails must have a common ground and must not back-feed the ESP32 USB supply.

APPENDIX B - DEMONSTRATION AND VALIDATION CHECKLIST

- Power on the prototype and observe ESP32/OLED startup.
- Confirm that both radar links report ONLINE when communicating.
- Set the potentiometer to low speed and verify 5 cm / 10 cm danger/warning limits.
- Test left zone at safe, warning and danger distances.
- Repeat the test for right and rear zones.
- Verify that an active danger zone overrides a safe reading elsewhere.
- Turn potentiometer to medium and high simulated speed bands.
- Confirm yellow and red buzzer patterns as risk changes.
- Connect a phone/laptop to the displayed access-point network and open the IP address.
- Verify browser telemetry, live OLED switching and return to IP mode after browser inactivity.
- Record any invalid echoes or anomalous reflections and adjust the sensor mounting as required.

Documentation note: Add actual prototype photographs, OLED snapshots and a short real sensor test log here before final institutional submission if those materials are required by the department. No measurements or photographs have been fabricated for this report.